

Kinematics of Uniform Circular Motion

SS ILP Physics (Intermediate), Week 6 Class 2



Outline

- Polar Coordinate System and Vectors in Polar Coordinates
- Uniform Circular Motion
- Non-Uniform Circular Motion
- Kinematics of Uniform Circular Motion

Reference:

University Physics with Modern Physics

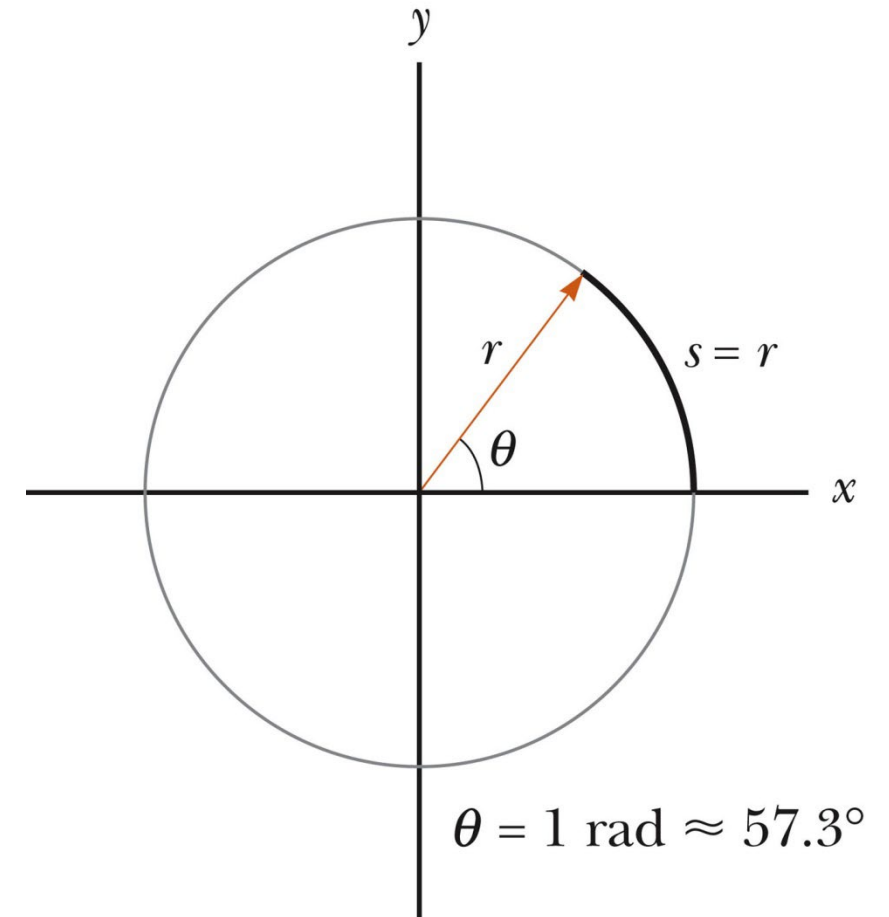
Angular Measurement: The Radian

- The radian is a unit of angular measure.
- The radian can be defined as the arc length s along a circle divided by the radius r ,

$$\theta = \frac{s}{r}$$

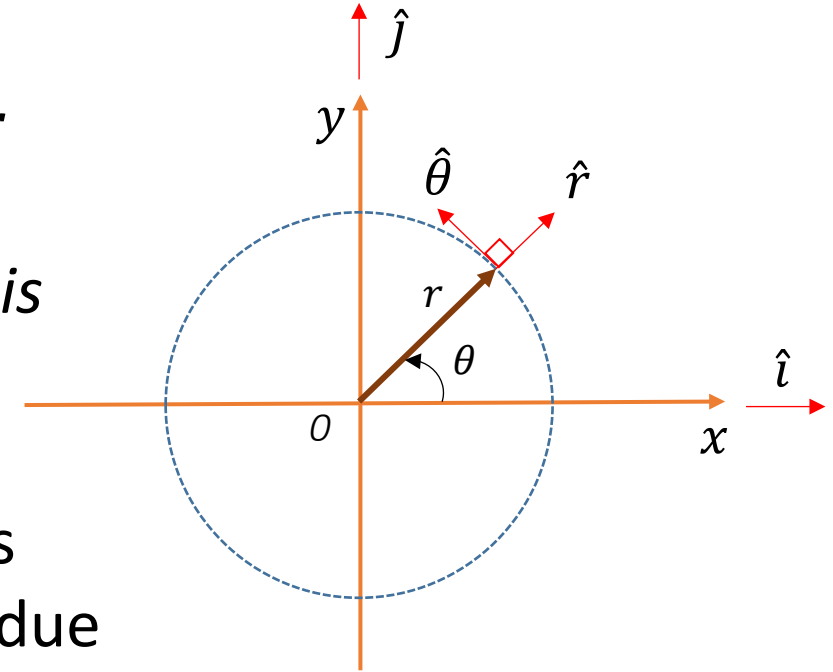
- Converting from degrees to radians,

$$\theta[\text{rad}] = \frac{\pi}{180^\circ} \times \theta[\text{degrees}]$$



Polar Coordinate System

- The position of point P is determined by the distance r from the origin O , and the angle θ measured with respect to x -axis (*conventionally anti-clock-wise angle is positive*)
- The coordinate system using distance r and angle θ is called polar coordinate system, it is mostly adopted (due to convenience) for studying of circular motion.
- The unit vectors $\hat{r}$ and $\hat{\theta}$ associated with the coordinate r and θ . $\hat{r}$ is radially pointing away from the origin, while $\hat{\theta}$ is tangential to the circular circumference (*in the direction of anti-clock-wise sense*) of radius r .

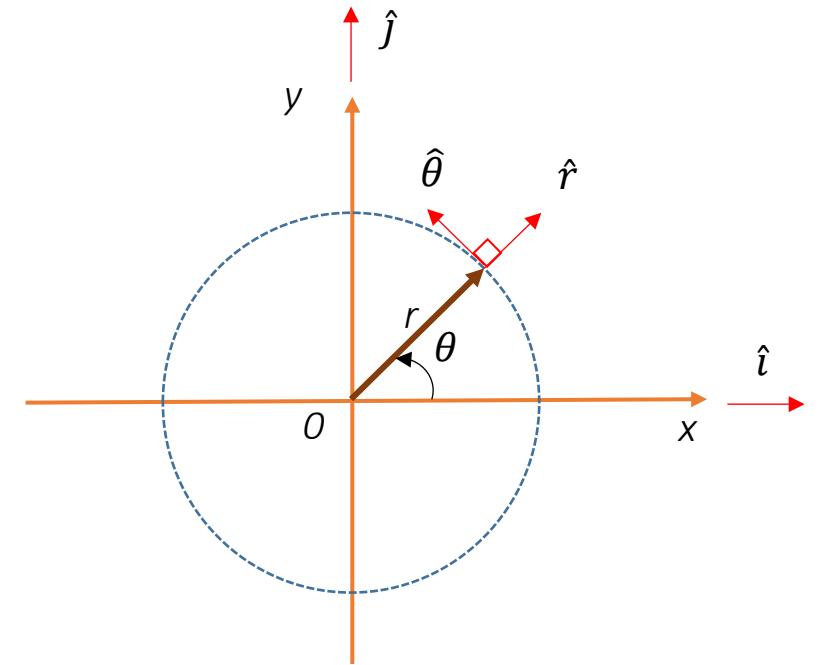


Relation Between Cartesian and Polar Coordinates

- The point P coordinate is given equivalently by (recall the trigonometric functions)

1. Cartesian coordinates (x, y)

2. Polar coordinates (r, θ)



The Cartesian description of the position vector $\vec{r}$ of point P is given by :

$$\vec{r} = x\hat{i} + y\hat{j} = r \cos(\theta) \hat{i} + r \sin(\theta) \hat{j}$$

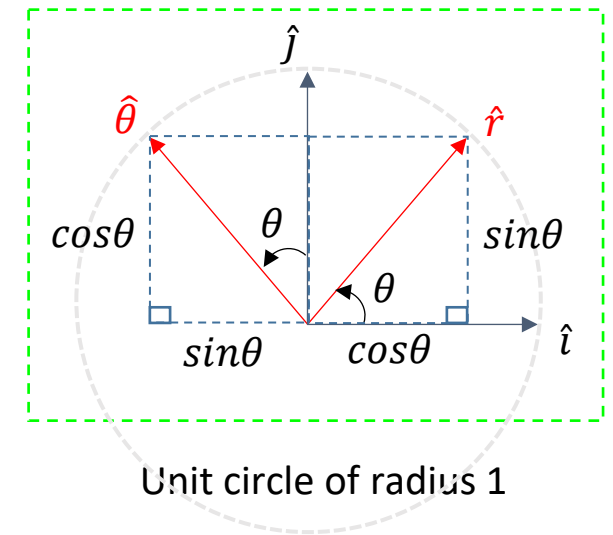
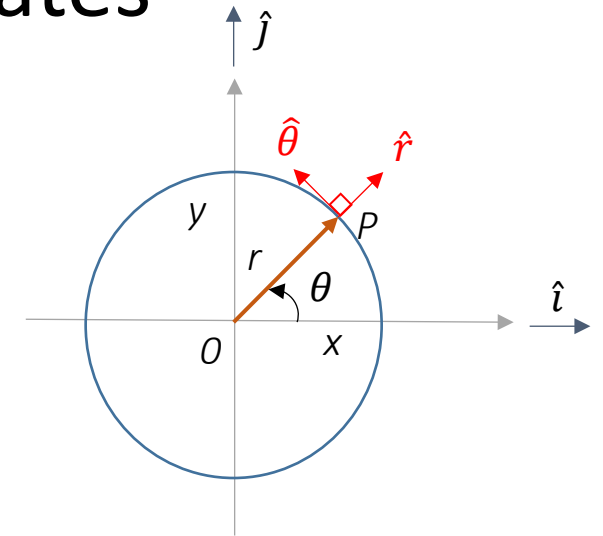
Relation Between Cartesian and Polar Coordinates

To get the relationship between the $\hat{i}, \hat{j}$ and $\hat{r}, \hat{\theta}$, we “bring down” both $\hat{r}$ and $\hat{\theta}$ vectors so that they no longer sit on the circumference of the circle but now on the origin of the coordinate system.

The polar description is given as follows:

- Position vector $\vec{r} = r\hat{r}$, no θ here since we are describing position so that $\hat{r} = \cos(\theta)\hat{i} + \sin(\theta)\hat{j}$

$$\rightarrow |\hat{r}|^2 = [\cos(\theta)]^2 + [\sin(\theta)]^2 = 1$$



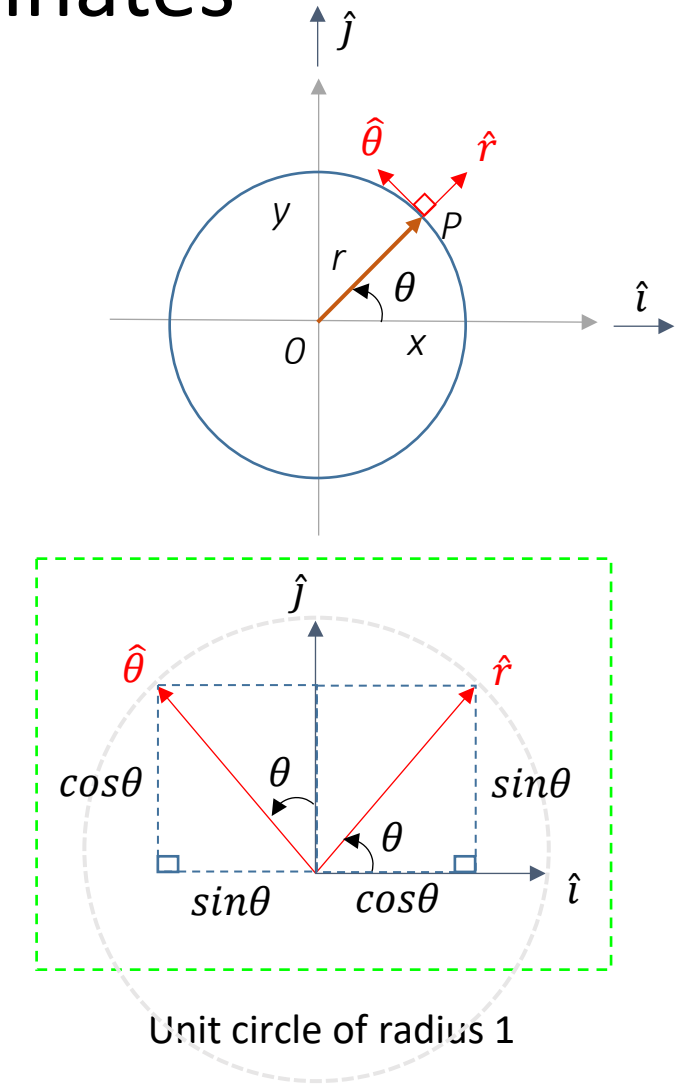
Relation Between Cartesian and Polar Coordinates

- The tangential vector $\hat{\theta}$ is, by definition, a vector that is perpendicular to $\hat{r}$.

Now we see that it is a vector that exists in the second quadrant. This means that the value of x is negative while the value of y is positive.

Using trigonometry, we see that $\hat{\theta} = -\sin(\theta)\hat{i} + \cos(\theta)\hat{j}$

$$\rightarrow |\hat{\theta}| = [-\sin(\theta)]^2 + [\cos(\theta)]^2 = 1$$



Concept Question

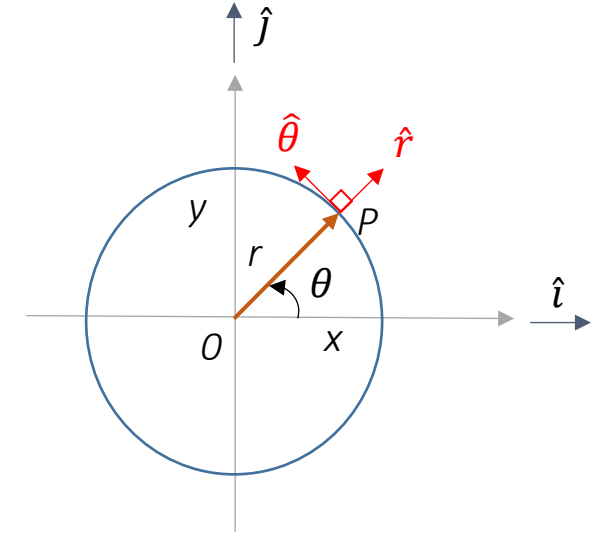
Find the Cartesian description of unit position vector $\hat{r}$ of point P .

a) $\hat{r} = r \cos(\theta) \hat{i} + r \sin(\theta) \hat{j}$

b) $\hat{r} = \cos(\theta) \hat{i} + \sin(\theta) \hat{j}$

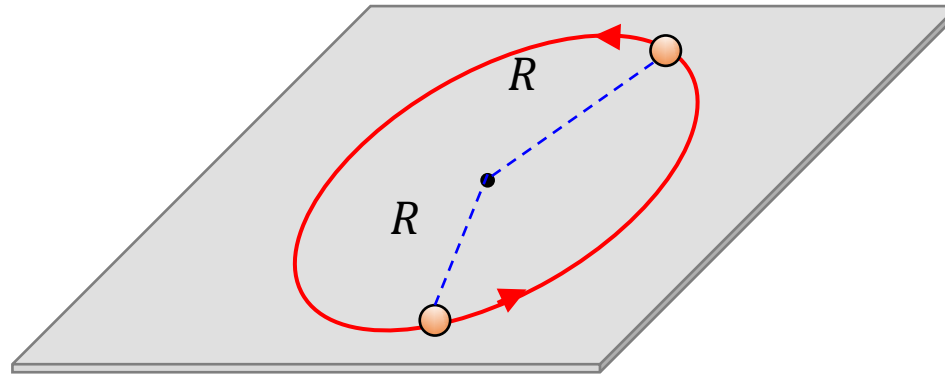
c) $\hat{r} = -\sin(\theta) \hat{i} + \cos(\theta) \hat{j}$

d) $\hat{r} = -r \sin(\theta) \hat{i} + r \cos(\theta) \hat{j}$



What is Circular Motion?

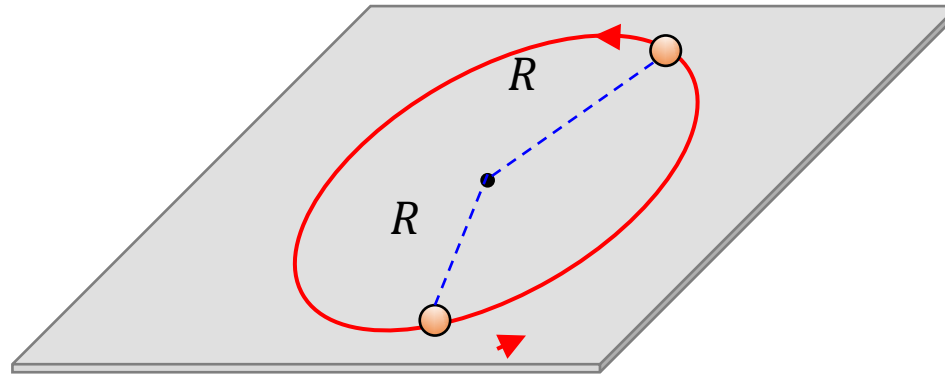
Circular motion is a special case of two-dimensional motion in which the object maintains a constant distance away from a fixed point.



What is Uniform Circular Motion?

Uniform Circular motion describes the motion of an object moving in a circle with constant speed.

If the speed is changing, the motion is called **non-uniform circular motion**.



H. D. Young and R. A. Freedman, University Physics (14th edition), p106-109

Some Basic Mathematics of Circular Motion

Recall that our position vector (in polar) is given by

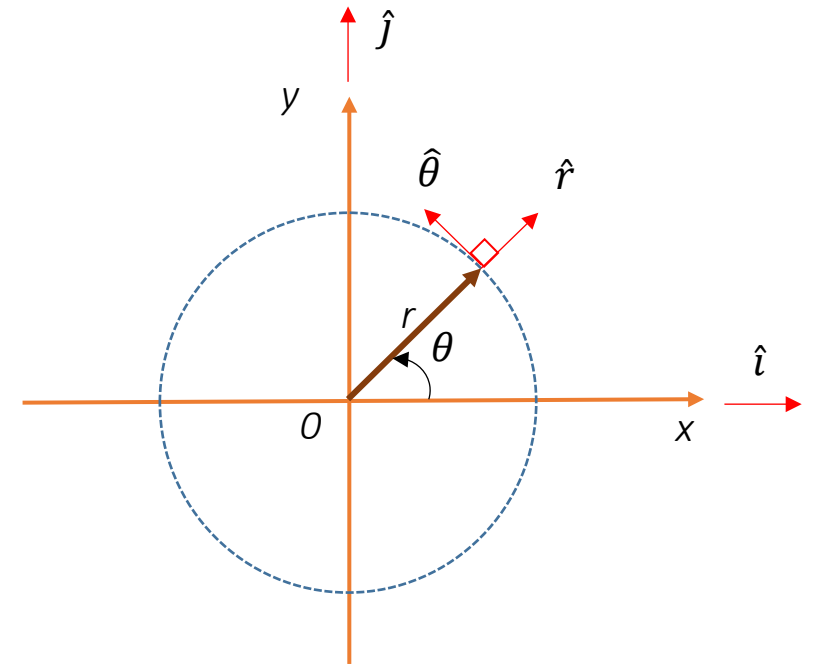
$$\vec{r} = r\hat{r} = r \cos(\theta) \hat{i} + r \sin(\theta) \hat{j}$$

As we undergo circular motion, the angle θ is going to change with time, i.e. $\theta(t)$.

So, we can express the angle θ as a function of time.

Consequently, our position vector is also a function in time.

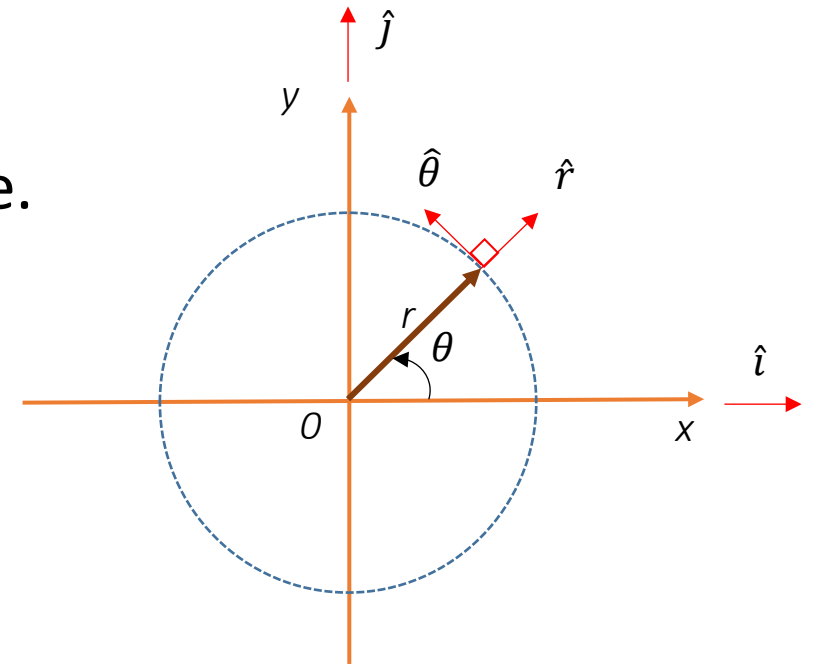
$$\vec{r}(t) = r[\hat{r}(t)] = r [\cos(\theta(t)) \hat{i} + \sin(\theta(t)) \hat{j}]$$



Some Basic Mathematics of Circular Motion

As we undergo circular motion, the unit vectors $\hat{r}$ and $\hat{\theta}$ (polar description) are also going to be functions of time.

This is because as the object moves along the circle, these unit vectors will change their orientation according to how the angle θ changes with time.



$$\hat{r}(t) = \cos(\theta(t)) \hat{i} + \sin(\theta(t)) \hat{j} \quad \rightarrow \quad |\hat{r}(t)| = 1$$

$$\hat{\theta}(t) = -\sin(\theta(t)) \hat{i} + \cos(\theta(t)) \hat{j} \quad \rightarrow \quad |\hat{\theta}(t)| = 1$$

Velocity and Acceleration for Uniform Circular Motion

We start with the definition of the radial vector in polar

coordinates: $\vec{r} = r[\cos(\theta(t))\hat{i} + \sin(\theta(t))\hat{j}] = r\hat{r}$

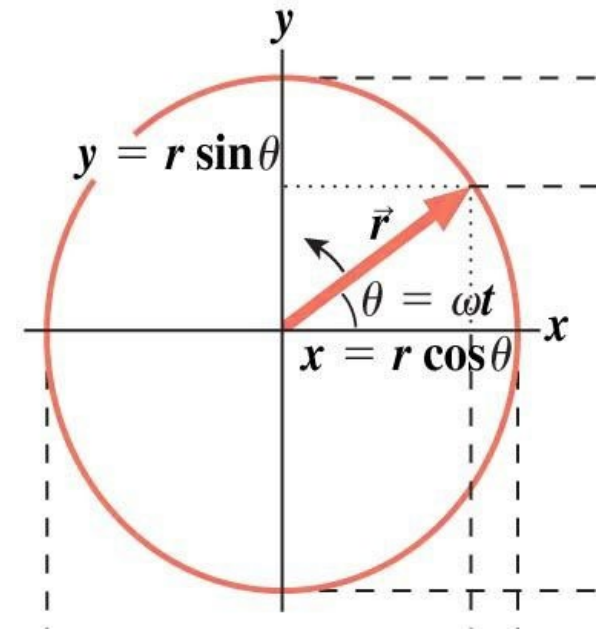
The tangential (linear) velocity is then given by

$$\vec{v} = \frac{d}{dt} \vec{r} = r \left[\frac{d}{dt} \{\cos(\theta(t))\} \hat{i} + \frac{d}{dt} \{\sin(\theta(t))\} \hat{j} \right]$$

Chain Rule

$$= r \left[-\sin(\theta(t)) \frac{d\theta}{dt} \hat{i} + \cos(\theta(t)) \frac{d\theta}{dt} \hat{j} \right] \rightarrow \vec{v} = r\omega [-\sin(\theta)\hat{i} + \cos(\theta)\hat{j}]$$

$$= r\omega\hat{\theta}$$



Velocity and Acceleration for Uniform Circular Motion

Angular speed (rad/s): $\omega = \frac{d\theta}{dt}$

Other unit: $1 \text{ rev/s} = 2\pi \text{ rad/s}$

Velocity and Acceleration for Uniform Circular Motion

The corresponding acceleration associated with **constant** tangential (linear) velocity is then given by

$$\begin{aligned}
 \vec{a} &= \frac{d}{dt} \vec{v} = r\omega \left[\frac{d}{dt} \{-\sin(\theta(t))\} \hat{i} + \frac{d}{dt} \{\cos(\theta(t))\} \hat{j} \right] \\
 &= r\omega \left[-\cos(\theta(t)) \frac{d\theta}{dt} \hat{i} - \sin(\theta(t)) \frac{d\theta}{dt} \hat{j} \right] = -r\omega^2 [\cos(\theta(t)) \hat{i} + \sin(\theta(t)) \hat{j}] \\
 &= -r\omega^2 \hat{r}(t)
 \end{aligned}$$

Chain Rule

Notice that though we have constant tangential velocity, circular motion allows us to have a non-zero acceleration. In the case of uniform circular motion, the acceleration is radial and points inwards. No tangential acceleration here.

Kinematics for Uniform Circular Motion

Position: $\vec{r}(t) = r [\cos(\theta(t)) \hat{i} + \sin(\theta(t)) \hat{j}] = r[\hat{r}(t)]$

Velocity: $\vec{v}(t) = r\omega [-\sin(\theta) \hat{i} + \cos(\theta) \hat{j}] = r\omega \hat{\theta}$

$$\omega = \frac{d\theta}{dt} \text{ and } v = r\omega$$

Acceleration: $\vec{a}(t) = -r\omega^2 [\hat{r}(t)] = -r\omega^2 [\cos(\theta(t)) \hat{i} + \sin(\theta(t)) \hat{j}]$

$$a = a_r = -r\omega^2 = -\frac{v^2}{r}$$

Concept Question

Find the acceleration of an object moving along a circle with a radius of a 0.5 m and at an angular velocity of 2 rad/s.

- (a) $2 \hat{r} \text{ rad/s}^2$
- (b) $-2 \hat{r} \text{ rad/s}^2$
- (c) $4 \hat{r} \text{ rad/s}^2$
- (d) $-4 \hat{r} \text{ rad/s}^2$

<https://openstax.org/books/physics/pages/6-2-uniform-circular-motion>

Concept Question

What is the centripetal acceleration felt by the passengers of a car moving at 6 m/s along a curve with radius 2.0 m?

<https://openstax.org/books/physics/pages/6-2-uniform-circular-motion>

Case Problem 1

A particle moves at an angular speed of 3×10^5 rad/s in a circle of radius $r = 2.0$ m. Choose $\theta_0 = 0$. Find

- (a) its position at any time,
- (b) its velocity at any time, and
- (c) its acceleration at any time.

[https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_\(OpenStax\)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_\(OpenStax\)/04%3A_Motion_in_Two_and_Three_Dimensions/4.05%3A_Uniform_Circular_Motion](https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_(OpenStax)/04%3A_Motion_in_Two_and_Three_Dimensions/4.05%3A_Uniform_Circular_Motion)

Case Problem 2

A proton has a speed of 5×10^6 m/s and is moving in a circle in the xy plane of radius $r = 0.175$ m. Find

- (a) its position at any time,
- (b) its velocity at any time, and
- (c) its acceleration at any time.

[https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_\(OpenStax\)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_\(OpenStax\)/04%3A_Motion_in_Two_and_Three_Dimensions/4.05%3A_Uniform_Circular_Motion](https://phys.libretexts.org/Bookshelves/University_Physics/University_Physics_(OpenStax)/Book%3A_University_Physics_I_-_Mechanics_Sound_Oscillations_and_Waves_(OpenStax)/04%3A_Motion_in_Two_and_Three_Dimensions/4.05%3A_Uniform_Circular_Motion)

Period of Uniform Circular Motion

- Time for one revolution, T is given by:

$$T = \frac{\text{circumference}}{\text{speed}} = \frac{2\pi r}{v}.$$

- Velocity is then

$$v = \frac{2\pi r}{T}.$$

- Acceleration can therefore be further written as:

$$a = a_r = \frac{v^2}{r} = \frac{4\pi^2 r}{T^2}.$$

Period of Uniform Circular Motion

- Frequency, the number of revolutions per second, is given by:

$$f = \frac{1}{T} = \frac{v}{2\pi r}.$$

- Velocity is then

$$v = 2\pi r f.$$

- Acceleration can therefore be further written as:

$$a = a_r = \frac{v^2}{r} = 4\pi^2 r f^2.$$

Concept Question

A 150-g ball at the end of a string is revolving uniformly in a horizontal circle of radius 0.600 m. The ball makes 2.00 revolutions in a second. What is its centripetal acceleration?

Concept Question

The moon's nearly circular orbit about the Earth has a radius of about 384,000 km and a period T of 27.3 days. Determine the acceleration of the Moon toward the Earth.

Concept Question

Passengers on a carnival ride move at constant speed in a horizontal circle of radius 5.0 m, making a complete circle in 4.0 s. What is their acceleration?

H. D. Young and R. A. Freedman, University Physics (14th edition), Example 3.12, p109